

1. Task Interface & Initialization



2. CREATE Reward-Engineering Agent

Observe

Outcome Dynamics Components Termination O_t

Reflect & Plan

Evidence tools Reflection LLM L1 Tune L2 Refactor L3 Redesign

Act, Validate & Remember

Reward edit Validate Memory R_{t+1}

3. Inner RL Environment

Policy training
(PPO in experiments)

Native-task
evaluation

Training record $\mathcal{D}_t$

4. Best Archive

Guarded
best reward

Final R^*

R_0

R_t / R_{t+1}

J_t